INCH-POUND

MIL-V-82808(OS)
29 December 1989
SUPERSEDING
WS 21190A
9 December 1986

MILITARY SPECIFICATION

VARNISH, PHENOLIC

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines one type of phenolic varnish, referred to herein as the varnish (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8010

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STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1963 Specific Gravity of Drying Oils, Varnishes, Resins, and Related Materials at 25 degrees/25 Degrees C, Test Method for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

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(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Chemical and physical properties. The chemical and physical properties of the varnish shall be as specified in table I.

TABLE I. Chemical and physical properties.

Property	Limits	
	Minimum	Maximum
Specific gravity, 25 degrees/ 25 degrees Celsius (°C)	1.07	1.10
Viscosity, centipoise, 25°C	180	1000
Resin solids, percent	60	64
pH	7.9	8.5

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3.3 **Safety.**

3.3.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.3.2 **Material Safety Data Sheet (MSDS).** The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.5).

3.4 **Age of material.** The material shall be stored at a maximum temperature of 40° Fahrenheit (F) and shipped within 30 days of manufacture date.

3.5 **Shelf life.** The varnish shall meet the requirements of this specification for a period of 6 months from the date of quality conformance inspection (see 4.4) when packaged as specified in section 5 and stored at 40°F maximum.

3.5.1 **Shelf life extension.** The shelf life may be extended for an additional 6 months provided the material, upon retest, conforms to the viscosity requirement specified in 3.2.

3.6 **Workmanship.** Workmanship shall be such that the varnish is uniform in appearance and free from visible contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the quality conformance inspections (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table II.

TABLE II. First article and quality conformance inspection.

Examination and tests	Requirement	Test Method
Visual examination	3.6 and section 5	4.5.1
Specific gravity test	3.2	4.5.2
Viscosity test	3.2	4.5.3
Resin solids test	3.2	4.5.4
pH test	3.2	4.5.5

4.4.1 **Sampling for inspection.**

4.4.1.1 **Lot.** For inspection purposes, a lot is defined as all the phenolic varnish subject to inspection at one time from one of the following:

a. A single batch (see 6.4.1)

b. A continuous process (see 6.4.2)

c. A uniform blend of batches or a uniform blend of portions from a continuous process.

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4.4.1.2 **Selection of sample containers.** Sample containers of the varnish shall be selected at random from each lot in accordance with table III for the inspections specified in 4.5.1 through 4.5.5. Sample material shall be selected as specified in 4.4.1.3.

TABLE III. Sampling for inspection.

Number of Containers in the lot	Number of sample containers to be selected for inspection
1 - 3	ALL
4 - 40	3
41 - 65	4
66 - 110	5

4.4.1.3 **Sample material.** Sufficient material shall be selected at random from each sample container to perform all of the inspections specified in 4.5.1 through 4.5.5.

4.4.1.4 **Noncompliance of samples.** Failure of one or more of the sample containers of varnish to meet all of the specified requirements shall be cause for rejection of the lot.

4.5 **Methods of inspection.** The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all inspections defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Each sample of varnish (see 4.4.1.3) shall be visually examined for conformance to 3.6.

4.5.2 **Specific gravity test.** Specific gravity of each sample shall be determined in accordance with ASTM D 1963, using a Hubbard pycnometer, and shall be recorded to the nearest 0.001 unit.

4.5.3 **Viscosity test.** Viscosity of each sample shall be determined in accordance with the following:

a. Draw sample by suction into calibrated Cannon-Fenske viscometer.

b. Condition viscometer and sample at 25.0 $\pm$ 0.1°C in a constant temperature bath.

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c. Draw sample up to starting point and measure efflux time between the two calibrated points.

d. Calculate viscosity as follows: Viscosity, centipoise = 0.997 TCS

Where: T = Average efflux time between calibrated viscometer points, seconds

C = Calibration constant for viscometer

S = Specific gravity of sample at 25/25°C

0.997 = Factor for conversion of specific gravity to density at 25°C

e. Record viscosity to nearest centipoise.

4.5.4 **Resin solids test.** The resin solids for each sample shall be determined in accordance with the following:

a. Preheat three marked drying dishes in oven at 135 ± 1°C for 3 hours ± 3 minutes.

b. Cool dishes in silica gel desiccator to room temperature and weigh to nearest 0.1 milligram (mg).

c. Weigh 3.0 ± 0.1 grams (g) to nearest 0.1 mg into each tared dried dish.

d. Place samples on one grid tray in drying oven. The samples shall be placed at the same level as the thermometer bulb and shall be grouped around the bulb.

e. Heat for 3 hours ± 3 minutes at 135 ± 1°C and transfer samples to silica gel desiccator. Cool to room temperature.

f. Reweigh samples to nearest 0.1 mg.

g. Calculate percent resin solids as follows:

$$\text{Percent resin solids} = \frac{W3 - W1}{W2 - W1} \times 100$$

Where: W1 = weight of dish, g

W2 = original total weight, g

W3 = final weight, g

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h. Record resin solids to nearest 0.1 percent. If any of the samples deviate more than 0.5 percent from average of determinations, test shall be repeated.

4.5.5 **pH test.** The pH for each sample shall be determined in accordance with the following:

a. Measure pH with pH meter previously calibrated with buffer solution.

b. Record pH to nearest 0.1 unit.

4.6 **Inspection of packaging.** The varnish containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The varnish shall be packaged to afford adequate protection against loss, contamination, deterioration and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.5. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178 or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

c. Material trade name and identification.

d. Net weight of contents.

e. Lot number and date of quality conformance inspection (see 4.4).

f. Shelf life and inspection/test date for extension (see 3.5 and 3.5.1). In addition, the marking shall include the following:

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"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 6 MONTH PERIOD PROVIDED THE VARNISH, UPON RETEST, CONFORMS TO THE VISCOSITY REQUIREMENT SPECIFIED IN MIL-V-82808."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The varnish specified herein is intended for use in Dual Thrust Rocket Motor nozzles and components, Mark 104, of the Standard Missile.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

c. Whether first article inspection is required (see 3.1 and 6.3).

d. Applicable contract provisions for Material Safety Data Sheet (see 3.3.2 and 6.5.

e. Special marking, if other than as specified in 5.2.2.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors on the material to be inspected. The material should be a first article sample and should consist of a quantity of the varnish sufficient to perform all of the inspections specified in 4.3. The first article sample should be packaged and marked as specified in section 5. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions.

6.4.1 Batch. A batch is that amount of phenolic varnish made as one unit in the manufacturing process.

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6.4.2 **Continuous process.** A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the material is accumulated for a lot.

6.5 **Material Safety Data Sheet (MSDS).** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR), Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed MDSDs prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.6 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mark 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 8010-N284)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-V-82808	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
VARNISH, PHENOLIC

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON
c. ADDRESS (*Include Zip Code*)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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